Igor Steinmacher, PhD

Associate Professor
Associate Director for Graduate Programs and Academic Affairs
School of Informatics, Computing and Cyber Systems · Northern Arizona University · Flagstaff, AZ, USA

igor.steinmacher@nau.edu · igor.pro.br · Google Scholar · ORCID · DBLP · Scopus

Highlights

- Currently supervising 3 Ph.D. students
- Successfully supervised 20+ undergraduate research students, 9 M.Sc. and 11 Ph.D. students (as supervisor and co-supervisor)
- Supervisor of Bianca Trinkenreich's PhD dissertation, winner of the 2024 ACM SIGSOFT Outstanding Dissertation Award
- Best Paper / Distinguished Paper Awards at CHASE@ICSE 2026, ESEM 2024, ICSE 2023, CONVERSATIONS 2023, ICSE-SEIS 2022, ICSE 2021, ICSME 2020, OpenSym 2020, ICSE-SEET 2019, SBES 2018
- Higher education instructor since 2005
- Secured more than USD 2.5M as lead PI of NSF grants
- Lead guest editor of the Thematic Series on Free and Open Source Software Development, SpringerOpen Journal of Internet Services and Applications (JISA)
- Guest editor of the IEEE Software special issue “20 Years of Open Source—Impact on Software Engineering Practice”
- Citation index h44 (Google Scholar)

Education

- **2015** — Ph.D. in Computer Science, University of São Paulo (USP), Brazil
- **2005** — M.Sc. in Computer Science, Federal University of Rio Grande do Sul (UFRGS), Brazil
- **2003** — B.S. in Computer Science, State University of Maringá (UEM), Brazil

Professional Experience

- **2025–current** — Associate Professor & Associate Director for Graduate Programs and Academic Affairs, Northern Arizona University (NAU), School of Informatics, Computing and Cyber Systems, USA
- **2018–2025** — Assistant Professor, Northern Arizona University (NAU), School of Informatics, Computing and Cyber Systems, USA (on leave of absence from Jan 2021 to Jul 2022)
- **2017–2018** — Post-doctoral Scholar, Northern Arizona University (NAU), USA
- **2009–2021** — Faculty Member, Federal University of Technology – Paraná (UTFPR), Department of Computing, Brazil (on leave of absence from Jan 2017 to Dec 2020)
- **2013–2014** — Visiting Scholar, University of California, Irvine (UCI), Department of Informatics, USA
- **2006–2009** — Project Manager, HSBC Global Technology, Brazil
- **2006–2009** — Lecturer, Faculdades Santa Cruz, Brazil
- **2005–2006** — Software Developer, CELEPAR – Data Processing Company of Paraná, Brazil

Grants

- **CommUnityBuddy** — a conversational agent supporting open source contributors. Alfred P. Sloan Foundation, 2026–current, lead PI, USD 100K.
- **NSF POSE: expanding the data.table ecosystem** — for efficient big data manipulation in R. 2023–2027, lead PI, USD 731K, with Marco Gerosa and Toby Hocking.
- **NSF HSI: a learning environment for an open-source contribution model** — OSS-Doorway. 2023–2027, lead PI, USD 498K, with Marco Gerosa.
- **NSF RETTL: learning software engineering by contributing to real projects with chatbot assistance**. 2023–2027, co-PI, USD 562K, with Marco Gerosa (PI).
- **NSF IUSE: scaffolding computational thinking in introductory CS through a conversational agent** — SIGIL. 2023–2026, lead PI, USD 405K, with Marco Gerosa.
- **NSF CHS: Gender-Inclusive Open Source through Gender-Inclusive Tools**. 2019–2023, lead PI, USD 528K.
- **Fostering open source communities by supporting newcomer developers**. FAPESP / MCTI CGI.br (Brazil), 2017, lead PI, ~BRL 230K, with Igor Wiese and Ivanilton Polato.

Awards and Honors

- **2026** — ACM SIGSOFT Distinguished Paper — CHASE@ICSE 2026: “Governance in Practice: How Open Source Projects Define and Document Roles”
- **2026** — NAU Award for Graduate Research Mentorship — Northern Arizona University Research and Creative Activity Awards
- **2025** — Most Promising Research Scholar — Northern Arizona University
- **2024** — Best Paper Award — ESEM 2024: “ChatGPT application in Systematic Literature Reviews in Software Engineering: an evaluation of its accuracy to support the selection activity”
- **2024** — ACM SIGSOFT Outstanding Dissertation Award — awarded to Bianca Trinkenreich, my PhD advisee
- **2023** — ACM SIGSOFT Distinguished Paper — ICSE 2023: “Do I Belong? Modeling Sense of Virtual Community Among Linux Kernel Contributors”
- **2023** — Best Paper Award — CONVERSATIONS 2023: “Anticipating User Needs: Insights from Design Fiction on Conversational Agents for Computational Thinking”
- **2023** — Distinguished Reviewer Award — ICSE 2023
- **2022** — Best Paper Award — ICSE-SEIS 2022: “An Empirical Investigation on the Challenges Faced by Women in the Software Industry: A Case Study”
- **2021** — ACM SIGSOFT Distinguished Paper — ICSE 2021: “What Makes a Great Maintainer of Open Source Projects?”
- **2021–2022** — Brazilian Research Productivity Fellowship (PQ)
- **2020** — IEEE TCSE Distinguished Paper — ICSME 2020: “Effects of Adopting Code Review Bots on Pull Requests to OSS Projects”
- **2020** — Best Paper Award — OpenSym 2020: “Recommending Tasks to Newcomers in OSS Projects: How Do Mentors Handle It?”
- **2020** — Honorable Mention — CSCW 2020: “Hidden Figures: Roles and Pathways of Successful OSS Contributors”
- **2019** — ACM SIGSOFT Distinguished Paper — ICSE-SEET 2019: “Training software engineers using open-source software: the students' perspective”
- **2018** — Best Paper Award — SBES 2018: “An empirical study on task documentation in software crowdsourcing: the case of the topcoder platform”

Publications

57 journal articles and 147 conference and workshop papers, from DBLP. ★ marks award-winning papers.

Journal Articles

- Daniel Coutinho, Breno Braga, Theo Canuto, Juliana Alves Pereira, Wesley K. G. Assunção, **Igor Steinmacher**, Marco Aurélio Gerosa, Alessandro Garcia. Leveraging large language models for sentiment analysis in GitHub pull request discussions. *Empir. Softw. Eng.* 31(5): 140 (2026).
- João Vitor Souza Rocha, Igor Scaliante Wiese, Ivanilton Polato, Marco Aurélio Graciotto Silva, Reginaldo Ré, **Igor Steinmacher**, Walter Takashi Nakamura. Investigating the potential of using worked examples to help resolve issues in a GitHub project. *J. Syst. Softw.* 236: 112810 (2026).
- Genildo Gomes, Fernando Nogueira, João Gustavo Kienen, Bernardo Scarambone, Desiree Scarambone, Leandro Galvão, Tanara Lauschner, **Igor Steinmacher**, Tayana Conte, Bruno Gadelha. FeelIt: A Tool for Expressing Real-Time Emotions and Its Influence on the Audience Experience. *J. Braz. Comput. Soc.* 32(1): 1812-1826 (2026).
- Zixuan Feng, Reed Milewicz, Emerson Murphy-Hill, Tyler Menezes, Alexander Serebrenik, **Igor Steinmacher**, Anita Sarma. Charting Uncertain Waters: A Socio-Technical Roadmap for Sustaining Open Source Communities in the Age of GenAI. *ACM Trans. Softw. Eng. Methodol.* 35(8): 214:1-214:21 (2026).
- Doris Amoakohene, Anirban Chetia, **Igor Steinmacher**, Toby Hocking. Asymptotic Benchmarking Using the atime package. *R J.* 18(1): 169-187 (2026).
- Zixuan Feng, Katie Kimura, Bianca Trinkenreich, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. Addressing OSS Community Managers' Challenges in Contributor Retention. *ACM Trans. Softw. Eng. Methodol.* 35(7): 187:1-187:38 (2026).
- Ítalo Santos, Kátia Romero Felizardo, **Igor Steinmacher**, Marco Aurélio Gerosa. Software solutions for newcomers' onboarding in software projects: A systematic literature review. *Inf. Softw. Technol.* 177: 107568 (2025).
- Márcia Lima, **Igor Steinmacher**, Denae Ford, Grace Vorreuter, Ludimila Gonçalves, Tayana Conte, Bruno Gadelha. How are discussions linked? A link analysis study on GitHub Discussions. *J. Syst. Softw.* 219: 112196 (2025).
- Bianca Trinkenreich, Marco Aurélio Gerosa, Anita Sarma, **Igor Steinmacher**. Guidelines for Cultivating a Sense of Belonging to Reduce Developer Burnout. *IEEE Softw.* 42(2): 84-91 (2025).
- Carolline Pena, Bruno Cartaxo, **Igor Steinmacher**, Deepika Badampudi, Deyvson da Silva, Williby da Silva Ferreira, Adauto Almeida, Fernando Kamei, Sérgio Soares. Comparing the Efficacy of Rapid Review With a Systematic Review in the Software Engineering Field. *J. Softw. Evol. Process.* 37(1) (2025).
- Gabriel Aracena, Kyle Luster, Fabio Santos, **Igor Steinmacher**, Marco Aurélio Gerosa. Applying large language models to issue classification: Revisiting with extended data and new models. *Sci. Comput. Program.* 246: 103333 (2025).
- Camila A. Paiva, Raquel Maximino, Frederico Paiva, Rafael Accetta Vieira, Nicole Espanha, João Felipe Pimentel, Igor Wiese, Marco Aurélio Gerosa, **Igor Steinmacher**, Leonardo Murta, Vanessa Braganholo. Analyzing the adoption of database management systems throughout the history of open source projects. *Empir. Softw. Eng.* 30(3): 71 (2025).
- Zixuan Feng, Katie Kimura, Bianca Trinkenreich, Anita Sarma, **Igor Steinmacher**. Guiding the way: A systematic literature review on mentoring practices in open source software projects. *Inf. Softw. Technol.* 171: 107470 (2024).
- Martin P. Robillard, Deeksha M. Arya, Neil A. Ernst, Jin L. C. Guo, Maxime Lamothe, Mathieu Nassif, Nicole Novielli, Alexander Serebrenik, **Igor Steinmacher**, Klaas-Jan Stol. Communicating Study Design Trade-offs in Software Engineering. *ACM Trans. Softw. Eng. Methodol.* 33(5): 112:1-112:10 (2024).

- Marco Aurélio Gerosa, Bianca Trinkenreich, **Igor Steinmacher**, Anita Sarma. Can AI serve as a substitute for human subjects in software engineering research? *Autom. Softw. Eng.* 31(1): 13 (2024).
- Simone de França Tonhão, Thelma Elita Colanzi, **Igor Steinmacher**. A portal to catalog worked examples extracted from open source software projects to support the teaching of Software Engineering. *J. Softw. Eng. Res. Dev.* 12(1) (2024).
- Fabio Santos, Joseph Vargovich, Bianca Trinkenreich, Ítalo Santos, Jacob Penney, Ricardo Britto, João Felipe Pimentel, Igor Wiese, **Igor Steinmacher**, Anita Sarma, Marco Aurélio Gerosa. Tag that issue: applying API-domain labels in issue tracking systems. *Empir. Softw. Eng.* 28(5): 116 (2023).
- Patrícia G. F. Matsubara, **Igor Steinmacher**, Bruno Gadelha, Tayana Conte. Much more than a prediction: Expert-based software effort estimation as a behavioral act. *Empir. Softw. Eng.* 28(4): 98 (2023).
- Márcia Lima, **Igor Steinmacher**, Denae Ford, Evangeline Liu, Grace Vorreuter, Tayana Conte, Bruno Gadelha. Looking for related posts on GitHub discussions. *PeerJ Comput. Sci.* 9: e1567 (2023).
- Mairieli Wessel, Andy Zaidman, Marco Aurélio Gerosa, **Igor Steinmacher**. Guidelines for Developing Bots for GitHub. *IEEE Softw.* 40(3): 72-79 (2023).
- **Igor Steinmacher**, Paul M. Clarke, Eray Tuzun, Ricardo Britto. Editorial: Machine learning, software process, and global software engineering. *J. Softw. Evol. Process.* 35(6) (2023).
- **Igor Steinmacher**, Paul M. Clarke, Eray Tuzun, Ricardo Britto. Editorial: Best papers of the 14th International Conference on Software and System Processes (ICSSP 2020) and 15th International Conference on Global Software Engineering (ICGSE 2020). *J. Softw. Evol. Process.* 35(5) (2023).
- Bianca Trinkenreich, Igor Wiese, Anita Sarma, Marco Aurélio Gerosa, **Igor Steinmacher**. Women's Participation in Open Source Software: A Survey of the Literature. *ACM Trans. Softw. Eng. Methodol.* 31(4): 81:1-81:37 (2022).
- Bianca Trinkenreich, Marco Aurélio Gerosa, **Igor Steinmacher**. Women in Open Source: We Need to Talk About It. *Computer* 55(12): 145-149 (2022).
- Fabio Calefato, Marco Aurélio Gerosa, Giuseppe Iaffaldano, Filippo Lanubile, **Igor Steinmacher**. Will you come back to contribute? Investigating the inactivity of OSS core developers in GitHub. *Empir. Softw. Eng.* 27(3): 76 (2022).
- **Igor Steinmacher**, Rafael Prikladnicki, Tayana Conte. Summary of the 12th Brazilian Conference on Software and the 1st Latin American School on Software Engineering. *ACM SIGSOFT Softw. Eng. Notes* 47(3): 28-31 (2022).
- Patrícia Gomes Fernandes Matsubara, Bruno F. Gadelha, **Igor Steinmacher**, Tayana Uchôa Conte. SEXTAMT: A systematic map to navigate the wide seas of factors affecting expert judgment software estimates. *J. Syst. Softw.* 185: 111148 (2022).
- Mairieli Santos Wessel, Alexander Serebrenik, Igor Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. Quality gatekeepers: investigating the effects of code review bots on pull request activities. *Empir. Softw. Eng.* 27(5): 108 (2022).
- Bianca Trinkenreich, Mariam Guizani, Igor Wiese, Tayana Conte, Marco Aurélio Gerosa, Anita Sarma, **Igor Steinmacher**. Pots of Gold at the End of the Rainbow: What is Success for Open Source Contributors? *IEEE Trans. Software Eng.* 48(10): 3940-3953 (2022).
- Hema Susmita Padala, Christopher J. Mendez, Felipe Fronchetti, **Igor Steinmacher**, Zoe Steine-Hanson, Claudia Hilderbrand, Amber Horvath, Charles Hill, Logan Simpson, Margaret Burnett, Marco Aurélio Gerosa, Anita Sarma. How Gender-Biased Tools Shape Newcomer Experiences in OSS Projects. *IEEE Trans. Software Eng.* 48(2): 241-259 (2022).
- Raula Gaikovina Kula, Christoph Treude, Hideaki Hata, Sebastian Baltes, **Igor Steinmacher**, Marco Aurélio Gerosa, Winifred Kula Amini. Challenges for Inclusion in Software Engineering: The Case of the Emerging Papua New Guinean Society. *IEEE Softw.* 39(3): 67-76 (2022).

- Mariam Guizani, Amreeta Chatterjee, Bianca Trinkenreich, Mary Evelyn May, Geraldine J. Noa-Guevara, Liam James Russell, Griselda G. Cuevas Zambrano, Daniel Izquierdo-Cortazar, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. The Long Road Ahead: Ongoing Challenges in Contributing to Large OSS Organizations and What to Do. *Proc. ACM Hum. Comput. Interact.* 5(CSCW2): 407:1-407:30 (2021).
- Flavia Bernardini, André Grützmann, **Igor Steinmacher**, Awdren L. Fontão, Davi Viana, Flávio E. A. Horita. Editorial Vol. 14 No. 3. *Braz. J. Inf. Syst.* 14(3): 1-4 (2021).
- Mairieli Santos Wessel, Igor Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. Don't Disturb Me: Challenges of Interacting with Software Bots on Open Source Software Projects. *Proc. ACM Hum. Comput. Interact.* 5(CSCW2): 301:1-301:21 (2021).
- **Igor Steinmacher**, Sogol Balali, Bianca Trinkenreich, Mariam Guizani, Daniel Izquierdo-Cortazar, Griselda G. Cuevas Zambrano, Marco Aurélio Gerosa, Anita Sarma. Being a Mentor in open source projects. *J. Internet Serv. Appl.* 12(1): 7 (2021).
- Paolo Tell, David Raffo, Liguo Huang, **Igor Steinmacher**, Ricardo Britto, Eray Tüzün, Paul M. Clarke. Summary of the 1st ICSSP-ICGSE Joint Event. *ACM SIGSOFT Softw. Eng. Notes* 45(4): 31-34 (2020).
- Bianca Trinkenreich, Mariam Guizani, Igor Wiese, Anita Sarma, **Igor Steinmacher**. Hidden Figures: Roles and Pathways of Successful OSS Contributors. *Proc. ACM Hum. Comput. Interact.* 4(CSCW2): 180:1-180:22 (2020). ★ Honorable Mention
- Jefferson De Oliveira Silva, Igor Wiese, Daniel M. Germán, Christoph Treude, Marco Aurélio Gerosa, **Igor Steinmacher**. Google summer of code: Student motivations and contributions. *J. Syst. Softw.* 162 (2020).
- Edson Oliveira, Eduardo Fernandes, **Igor Steinmacher**, Marco Cristo, Tayana Conte, Alessandro Garcia. Code and commit metrics of developer productivity: a study on team leaders perceptions. *Empir. Softw. Eng.* 25(4): 2519-2549 (2020).
- Gregorio Robles, **Igor Steinmacher**, Paul Adams, Christoph Treude. Twenty Years of Open Source Software: From Skepticism to Mainstream. *IEEE Softw.* 36(6): 12-15 (2019).
- Dan Sholler, **Igor Steinmacher**, Denae Ford, Mara Averick, Mike Hoye, Greg Wilson. Ten simple rules for helping newcomers become contributors to open projects. *PLoS Comput. Biol.* 15(9) (2019).
- Williamson Silva, **Igor Steinmacher**, Tayana Conte. Students' and Instructors' Perceptions of Five Different Active Learning Strategies Used to Teach Software Modeling. *IEEE Access* 7: 184063-184077 (2019).
- Igor Scaliante Wiese, Rodrigo Takashi Kuroda, **Igor Steinmacher**, Gustavo Ansaldi Oliva, Reginaldo Ré, Christoph Treude, Marco Aurélio Gerosa. Pieces of contextual information suitable for predicting co-changes? An empirical study. *Softw. Qual. J.* 27(4): 1481-1503 (2019).
- **Igor Steinmacher**, Marco Aurélio Gerosa, Tayana Uchôa Conte, David F. Redmiles. Overcoming Social Barriers When Contributing to Open Source Software Projects. *Comput. Support. Cooperative Work.* 28(1-2): 247-290 (2019).
- **Igor Steinmacher**, Christoph Treude, Marco Aurélio Gerosa. Let Me In: Guidelines for the Successful Onboarding of Newcomers to Open Source Projects. *IEEE Softw.* 36(4): 41-49 (2019).
- Luis Felipe Dias, **Igor Steinmacher**, Gustavo Pinto. Who drives company-owned OSS projects: internal or external members? *J. Braz. Comput. Soc.* 24(1): 16:1-16:17 (2018).
- Mairieli Santos Wessel, Bruno Mendes de Souza, **Igor Steinmacher**, Igor Scaliante Wiese, Ivanilton Polato, Ana Paula Chaves, Marco Aurélio Gerosa. The Power of Bots: Characterizing and Understanding Bots in OSS Projects. *Proc. ACM Hum. Comput. Interact.* 2(CSCW): 182:1-182:19 (2018).
- Gustavo Pinto, **Igor Steinmacher**, Luiz Felipe Dias, Marco Aurélio Gerosa. On the challenges of open-sourcing proprietary software projects. *Empir. Softw. Eng.* 23(6): 3221-3247 (2018).
- Sogol Balali, **Igor Steinmacher**, Umayal Annamalai, Anita Sarma, Marco Aurélio Gerosa. Newcomers' Barriers. . . Is That All? An Analysis of Mentors' and Newcomers' Barriers in OSS Projects. *Comput. Support. Cooperative Work.* 27(3-6): 679-714 (2018).

- Anderson Bergamini de Neira, **Igor Steinmacher**, Igor Scaliante Wiese. Characterizing the hyperspecialists in the context of crowdsourcing software development. *J. Braz. Comput. Soc.* 24(1): 17:1-17:16 (2018).
- Igor Scaliante Wiese, Reginaldo Ré, **Igor Steinmacher**, Rodrigo Takashi Kuroda, Gustavo Ansaldi Oliva, Christoph Treude, Marco Aurélio Gerosa. Using contextual information to predict co-changes. *J. Syst. Softw.* 128: 220-235 (2017).
- **Igor Steinmacher**, Gregorio Robles, Brian Fitzgerald, Anthony I. Wasserman. Free and open source software development: the end of the teenage years. *J. Internet Serv. Appl.* 8(1): 17:1-17:4 (2017).
- Alexandre Lazaretti Zanatta, **Igor Steinmacher**, Letícia S. Machado, Cleidson R. B. de Souza, Rafael Prikladnicki. Barriers Faced by Newcomers to Software-Crowdsourcing Projects. *IEEE Softw.* 34(2): 37-43 (2017).
- **Igor Steinmacher**, Marco Aurélio Graciotto Silva, Marco Aurélio Gerosa, David F. Redmiles. A systematic literature review on the barriers faced by newcomers to open source software projects. *Inf. Softw. Technol.* 59: 67-85 (2015).
- Igor Scaliante Wiese, Douglas Nassif Junior, Reginaldo Ré, **Igor Steinmacher**, Marco Aurélio Gerosa. Comparing communication and development networks for predicting file change proneness: An exploratory study considering process and social metrics. *Electron. Commun. Eur. Assoc. Softw. Sci. Technol.* 65 (2014).
- **Igor Steinmacher**, Ana Paula Chaves, Marco Aurélio Gerosa. Awareness Support in Distributed Software Development: A Systematic Review and Mapping of the Literature. *Comput. Support. Cooperative Work.* 22(2-3): 113-158 (2013).
- Ana Paula Chaves, **Igor Steinmacher**, Gislaine Camila Lapasini Leal, Elisa Hatsue Moriya Huzita, Alberto B. Biasão. OntoDiSENv1: an Ontology to Support Global Software Development. *CLEI Electron. J.* 14(2) (2011).

Conference and Workshop Papers

- Nathan Cassee, Sankarsh Ravi, Ítalo Santos, **Igor Steinmacher**, Alexander Serebrenik. What barriers do students experience when trying to contribute to Open Source Software projects? *SEET@ICSE* , 151-161 (2026).
- Karl R. Wurst, Erik Falor, Avni Khatri, **Igor Steinmacher**. Ways to Make Git Easier to Teach. *SIGCSE* , 1627-1628 (2026).
- Mika Mäntylä, Patrícia Matsubara, Kátia Romero Felizardo, Miikka Kuutila, Marco Aurélio Gerosa, Savio Sousa Sampaio, Tayana Conte, **Igor Steinmacher**. Understanding LLMs in Title-Abstract Screening: From Disagreements to Recommendations. *SEAA* , 226-242 (2026).
- Sadia Afroz, Zixuan Feng, Tyler Menezes, Katie Kimura, Bianca Trinkenreich, **Igor Steinmacher**, Anita Sarma. The Fast and Spurious: Developer Productivity with GenAI. *SIGSOFT FSE Companion* , 1832-1841 (2026).
- Pawan Acharya, Tayana Conte, Jacob Penney, Pedro Oliveira, Misan Paul Etchie, Jared Duval, Bruno Gadelha, Marco Aurélio Gerosa, **Igor Steinmacher**. Shh! Don't Tell Them I Use GenAI: Exploring How Participants Use Generative AI in Hackathons. *SIGSOFT FSE Companion* , 1842-1850 (2026).
- Pedro Oliveira, Tayana Conte, Marco Aurélio Gerosa, **Igor Steinmacher**. Governance in Practice: How Open Source Projects Define and Document Roles. *CHASE@ICSE* , 255-266 (2026). ★ ACM SIGSOFT Distinguished Paper
- Raula Gaikovina Kula, Christoph Treude, Xing Hu, Sebastian Baltes, Earl T. Barr, Kelly Blincoe, Fabio Calefato, Junjie Chen, Marc Cheong, Youmei Fan, Daniel M. Germán, Marco Aurélio Gerosa, Jin L. C. Guo, Shinpei Hayashi, Robert Hirschfeld, Reid Holmes, Yintong Huo, Takashi Kobayashi, Michele Lanza, Zhongxin Liu, Olivier Nourry, Nicole Novielli, Denys Poshyvanyk, Shinobu Saito, Kazumasa Shimari, **Igor Steinmacher**, Mairieli Wessel, Markus Wagner, Annie Vella, Laurie Williams, Xin Xia. Forecasting Developer Environments with GenAI: A Research Perspective. *IDE@ICSE* , 1-5 (2026).

- Ashish Amresh, Siddharth Subramanian, **Igor Steinmacher**. Evaluating the Effect of Generative AI Content in Common Game Development Software Architectures. *GAS@ICSE* , 47-54 (2026).
- Rudrajit Choudhuri, Bianca Trinkenreich, Rahul Pandita, Eirini Kalliamvakou, **Igor Steinmacher**, Marco Aurélio Gerosa, Christopher Sanchez, Anita Sarma. What Guides Our Choices? Modeling Developers' Trust and Behavioral Intentions Towards Genai. *ICSE* , 1691-1703 (2025).
- Reine Santos, Gabriel da Silva Freitas, **Igor Steinmacher**, Tayana Conte, Ana Carolina Oran, Bruno Gadelha. User Stories: Does ChatGPT Do It Better? *ICEIS* , 47-58 (2025).
- Genildo Gomes, Paulo Silva, Sâmila Magalhães, Leandro Galvão, Tanara Lauschner, **Igor Steinmacher**, Tayana Conte, Bruno Gadelha. Understanding the Student Experience Better: Analyzing Remote Learning Tools Through Engagement, Participation, and Liveness Concepts. *CSEDU* , 958-965 (2025).
- Jacob Penney, Pawan Acharya, Peter Hilbert, Priyanka Parekh, Anita Sarma, **Igor Steinmacher**, Marco Aurélio Gerosa. Understanding Programming Students' Help-Seeking Preferences in the Era of Generative AI. *CompEd* , 15-21 (2025).
- Misan Paul Etchie, Hunter M. Beach, Kátia Romero Felizardo, **Igor Steinmacher**. Towards a Dataset to Assess Conversational Agent's Efficacy in Mentoring Newcomer Developers. *BotSE@ICSE* , 48-52 (2025).
- Zixuan Feng, **Igor Steinmacher**, Marco Aurélio Gerosa, Tyler Menezes, Alexander Serebrenik, Reed Milewicz, Anita Sarma. The Multifaceted Nature of Mentoring in OSS: Strategies, Qualities, and Ideal Outcomes. *CHASE@ICSE* , 203-214 (2025).
- Benjamin C. Carter, Jonathan Rivas Contreras, Carlos A. Llanes Villegas, Pawan Acharya, Jack Utzerath, Adonijah O. Farner, Hunter Jenkins, Dylan Johnson, Jacob Penney, **Igor Steinmacher**, Marco Aurélio Gerosa, Fabio Santos. SkillScope: A Tool to Predict Fine-Grained Skills Needed to Solve Issues on GitHub. *NLBSE@ICSE* , 9-12 (2025).
- Daniel Coutinho, Juliana Alves Pereira, Breno Braga Neves, João Lucas Correia, Caio Barbosa, Wesley K. G. Assunção, **Igor Steinmacher**, Marco Aurélio Gerosa, Augusto Baffa, Alessandro Garcia. PRemo: A Dataset of Emotions Found on Pull Request Discussions. *SBES* , 326-336 (2025).
- Jacob Penney, Pawan Acharya, Peter Hilbert, Priyanka Parekh, Anita Sarma, **Igor Steinmacher**, Marco Aurélio Gerosa. Outcomes, Perceptions, and Interaction Strategies of Novice Programmers Studying with ChatGPT. *CUI* , 73:1-73:15 (2025).
- Ítalo Santos, Kátia Romero Felizardo, Anita Sarma, **Igor Steinmacher**, Marco Aurélio Gerosa. Ossdoorway: A Gamified Environment to Scaffold Student Contributions to Open Source Software. *CSEE&T* , 325-335 (2025).
- Kátia Romero Felizardo, Anderson Deizepe, Daniel Coutinho, Genildo Gomes, Maria A. C. Meireles, Marco Aurélio Gerosa, **Igor Steinmacher**. On the Difficulties of Conducting and Replicating Systematic Literature Reviews Studies Using LLMs in Software Engineering. *WSESE@ICSE* , 20-23 (2025).
- Andressa S. S. Medeiros, Thelma Elita Colanzi, Igor Scaliante Wiese, Igor Fabio Steinmacher. O uso de exemplos trabalhados para facilitar a contribuição de novatos em projetos de software livre. *SBES* , 459-469 (2025).
- Bianca Trinkenreich, Zixuan Feng, Rudrajit Choudhuri, Marco Aurélio Gerosa, Anita Sarma, **Igor Steinmacher**. Investigating the Impact of Interpersonal Challenges on Feeling Welcome in OSS. *ICSE* , 717-729 (2025).
- Ana Maciel, Mairieli Wessel, Igor Wiese, **Igor Steinmacher**. Investigating the Impact of GitHub Discussions on Maintainers' Workload and Community Dynamics. *SBES* , 37-47 (2025).
- Rudrajit Choudhuri, Ambareesh Ramakrishnan, Amreeta Chatterjee, Bianca Trinkenreich, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. Insights from the Frontline: GenAI Utilization Among Software Engineering Students. *CSEE&T* , 1-12 (2025).
- Igor Scaliante Wiese, Jasmine Boyer, Ethan Rasgorshek, Gustavo Pinto, Marco Aurélio Gerosa, **Igor Steinmacher**, Bonita Sharif. How Developers Make Decisions When Choosing Issues and Reviewing Code: An Eye Tracking GitHub Study. *ETRA* , 2:1-2:7 (2025).

- Ítalo Santos, Kátia Romero Felizardo, **Igor Steinmacher**, Marco Aurélio Gerosa. Great Power Brings Great Responsibility: Personalizing Conversational AI for Diverse Problem-Solvers. *CHASE@ICSE* , 93-95 (2025).
- Pedro Oliveira, Doris Amoakohene, Toby Hocking, Marco Aurélio Gerosa, **Igor Steinmacher**. Governance Matters: Lessons From Restructuring the Data.Table OSS Project. *ICSME* , 1-12 (2025).
- Ítalo Santos, Kátia Romero Felizardo, Bianca Trinkenreich, Daniel M. Germán, **Igor Steinmacher**, Marco Aurélio Gerosa. Exploring the Untapped: Student Perceptions and Participation in OSS. *SIGSOFT FSE Companion* , 859-870 (2025).
- Bruno Gadelha, Thiago Queiroz, Cleidson R. B. de Souza, Tayana Conte, **Igor Steinmacher**. Enhancing User-centered Design with Large Language Models. *VL/HCC* , 367-372 (2025).
- Maria Alcimar Costa Meireles, Eriky Rodrigues, **Igor Steinmacher**, José Carlos Maldonado, Tayana Conte. Design Thinking as a Toolbox for Requirements Engineering Education: Lessons from the Classroom. *SBES* , 533-543 (2025).
- Marcelo Costalonga, Bianca Minetto Napoleão, Maria Teresa Baldassarre, Kátia Romero Felizardo, **Igor Steinmacher**, Marcos Kalinowski. Can Machine Learning Support the Selection of Studies for Systematic Literature Review Updates? *WSESE@ICSE* , 56-63 (2025).
- Kátia Romero Felizardo, Francisco Carlos M. Souza, Alinne Cristinne Corrêa Souza, Bianca Minetto Napoleão, **Igor Steinmacher**, Marco Aurélio Gerosa. Assessing Diversity in Creating Seed Set for Snowballing Search for Systematic Literature Review in Software Engineering. *ESEM* , 1-11 (2025).
- Thiago Queiroz, Gretchen Torres de Macedo, **Igor Steinmacher**, Tayana Conte, Ana Carolina Oran, Bruno Gadelha. An Approach to Assist the Design of User-Centered Solutions based on Generative AI. *SBQS* , 215-225 (2025).
- Reine Santos, **Igor Steinmacher**, Tayana Conte, Ana Carolina Oran, Bruno Gadelha. Adoption of LLMs in Requirements Engineering: What Practitioners Are Worried About? *SBQS* , 248-258 (2025).
- Reine Santos, **Igor Steinmacher**, Tayana Conte, Ana Carolina Oran, Bruno Gadelha. AI-Generated User Stories: Are They Good Enough? *SBES* , 741-747 (2025).
- Isabella Macedo, Maria Alcimar Costa Meireles, Hannah Barreto, **Igor Steinmacher**, José Carlos Maldonado, Bruno Gadelha, Tayana Conte. AI in the Design Thinking Toolbox: What to Recommend, When, and Why. *SBES* , 755-761 (2025).
- João Lucas Correia, Morgan C. Nicholson, Daniel Coutinho, Caio Barbosa, Marco Castelluccio, Marco Aurélio Gerosa, Alessandro F. Garcia, **Igor Steinmacher**. Unveiling the Potential of a Conversational Agent in Developer Support: Insights from Mozilla's PDF.js Project. *AIware* (2024).
- Bianca Trinkenreich, Marco Aurélio Gerosa, **Igor Steinmacher**. Unraveling the Drivers of Sense of Belonging in Software Delivery Teams: Insights from a Large-Scale Survey. *ICSE* , 190:1-190:12 (2024).
- Saori Costa, Matheus Paixão, **Igor Steinmacher**, Pamella Soares, Allysson Allex Araújo, Jerffeson Souza. Sociotechnical Dynamics in Open Source Smart Contract Repositories: An Exploratory Data Analysis of Curated High Market Value Projects. *PROMISE* , 22-31 (2024).
- João Lucas Correia, Daniel Coutinho, Alessandro Garcia, Rafael Maiani de Mello, Caio Barbosa, Anderson Oliveira, Wesley K. G. Assunção, Juliana Alves Pereira, **Igor Steinmacher**, Marco Aurélio Gerosa, Jairo Souza, Johny Arriel. On the Investigation of Exception Pull Request Characteristics: Exploring the Apache Ecosystem. *SCAM* , 143-153 (2024).
- Emily Judith Arteaga Garcia, João Felipe Nicolaci Pimentel, Zixuan Feng, Marco Aurélio Gerosa, **Igor Steinmacher**, Anita Sarma. How to Support ML End-User Programmers through a Conversational Agent. *ICSE* , 53:1-53:12 (2024).
- Rudrajit Choudhuri, Dylan Liu, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. How Far Are We? The Triumphs and Trials of Generative AI in Learning Software Engineering. *ICSE* , 184:1-184:13 (2024).
- Ítalo Santos, Kátia Romero Felizardo, Marco Aurélio Gerosa, **Igor Steinmacher**. Game Elements to Engage Students Learning the Open Source Software Contribution Process. *VL/HCC* , 59-70 (2024).

- Gustavo Pinto, Cleidson R. B. de Souza, Thayssa A. da Rocha, **Igor Steinmacher**, Alberto de Souza, Edward Monteiro. Developer Experiences with a Contextualized AI Coding Assistant: Usability, Expectations, and Outcomes. *CAIN* , 81-91 (2024).
- Kátia Romero Felizardo, **Igor Steinmacher**, Márcia Sampaio Lima, Anderson Deizepe, Tayana Uchôa Conte, Monalessa Perini Barcellos. Data extraction for systematic mapping study using a large language model - a proof-of-concept study in software engineering. *ESEM* , 407-413 (2024).
- Kátia Romero Felizardo, Márcia Sampaio Lima, Anderson Deizepe, Tayana Uchôa Conte, **Igor Steinmacher**. ChatGPT application in Systematic Literature Reviews in Software Engineering: an evaluation of its accuracy to support the selection activity. *ESEM* , 25-36 (2024). ★ Best Paper Award
- **Igor Steinmacher**, Jacob McAuley Penney, Kátia Romero Felizardo, Alessandro F. Garcia, Marco Aurélio Gerosa. Can ChatGPT emulate humans in software engineering surveys? *ESEM* , 414-419 (2024).
- Gabriel Aracena, Kyle Luster, Fabio Santos, **Igor Steinmacher**, Marco Aurélio Gerosa. Applying Large Language Models to Issue Classification. *NLBSE@ICSE* , 57-60 (2024).
- Nicole Davila, Igor Wiese, **Igor Steinmacher**, Lucas Lucio da Silva, Andre Kawamoto, Gilson Jose Peres Favaro, Ingrid Nunes. An Industry Case Study on Adoption of AI-based Programming Assistants. *ICSE-SEIP* , 92-102 (2024).
- Daniel Coutinho, Luisa Cito, Maria Vitória Lima, Beatriz Arantes, Juliana Alves Pereira, Johny Arriel, João Godinho, Vinicius Martins, Paulo Vítor C. F. Libório, Leonardo Pedrosa Leite, Alessandro Garcia, Wesley K. G. Assunção, **Igor Steinmacher**, Augusto Baffa, Baldoino Fonseca. "Looks Good To Me ;-)": Assessing Sentiment Analysis Tools for Pull Request Discussions. *EASE* , 211-221 (2024).
- Fabio Santos, Jacob Penney, João Felipe Pimentel, Igor Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. Tell Me Who Are You Talking to and I Will Tell You What Issues Need Your Skills. *MSR* , 611-623 (2023).
- Patrícia Gomes Fernandes Matsubara, **Igor Steinmacher**, Bruno Gadelha, Tayana Conte. SwEDeL: Software Estimates' Defense Lenses Designed from Negotiation Methods. *SBQS* , 352-354 (2023).
- Mariam Guizani, Aileen Abril Castro-Guzman, Anita Sarma, **Igor Steinmacher**. Rules of Engagement: Why and How Companies Participate in OSS. *ICSE* , 2617-2629 (2023).
- Patrícia G. F. Matsubara, **Igor Steinmacher**, Bruno Gadelha, Tayana Conte. Moving on from the Software Engineers' Gambit: An Approach to Support the Defense of Software Effort Estimates. *ICSE* , 703-715 (2023).
- Joseph Vargovich, Fabio Santos, Jacob Penney, Marco Aurélio Gerosa, **Igor Steinmacher**. GiveMeLabeledIssues: An Open Source Issue Recommendation System. *MSR* , 402-406 (2023).
- Simone de França Tonhão, Marcelo Shigenaga, Júlio Budiski Herculani, Andressa S. S. Medeiros, Aline M. M. M. Amaral, Williamson Silva, Thelma Elita Colanzi, **Igor Steinmacher**. Gamification in Software Engineering Education: a Tertiary Study. *SBES* , 358-367 (2023).
- Bianca Trinkenreich, Klaas-Jan Stol, Anita Sarma, Daniel M. Germán, Marco Aurélio Gerosa, **Igor Steinmacher**. Do I Belong? Modeling Sense of Virtual Community Among Linux Kernel Contributors. *ICSE* , 319-331 (2023). ★ ACM SIGSOFT Distinguished Paper
- Felipe Fronchetti, David C. Shepherd, Igor Wiese, Christoph Treude, Marco Aurélio Gerosa, **Igor Steinmacher**. Do CONTRIBUTING Files Provide Information about OSS Newcomers' Onboarding Barriers? *ESEC/SIGSOFT FSE* , 16-28 (2023).
- Ítalo Santos, João Felipe Pimentel, Igor Wiese, **Igor Steinmacher**, Anita Sarma, Marco Aurélio Gerosa. Designing for Cognitive Diversity: Improving the GitHub Experience for Newcomers. *ICSE-SEIS* , 1-12 (2023).
- Larissa Salerno, Simone de França Tonhão, **Igor Steinmacher**, Christoph Treude. Barriers and Self-Efficacy: A Large-Scale Study on the Impact of OSS Courses on Student Perceptions. *ITiCSE* , 320-326 (2023).
- Jacob Penney, João Felipe Pimentel, **Igor Steinmacher**, Marco Aurélio Gerosa. Anticipating User Needs: Insights from Design Fiction on Conversational Agents for Computational Thinking. *CONVERSATIONS* , 204-219 (2023). ★ Best Paper Award

- Bianca Trinkenreich, Klaas-Jan Stol, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma, Marcelo Lara, Michael Feathers, Nicholas Ross, Kevin Bishop. A Model for Understanding and Reducing Developer Burnout. *ICSE-SEIP* , 48-60 (2023).
- Carlos Danilo Luz, Edson Oliveira Jr., **Igor Steinmacher**. A Conceptual Model to Support Teaching of Software Engineering Controlled (Quasi-)Experiments. *SBES* , 236-245 (2023).
- Eric Ribeiro, Ronan Nascimento, **Igor Steinmacher**, Laerte Xavier, Marco Aurélio Gerosa, Hugo de Paula, Mairieli Wessel. Together or Apart? Investigating a mediator bot to aggregate bot's comments on pull requests. *ICSME* , 434-438 (2022).
- Patrícia G. F. Matsubara, **Igor Steinmacher**, Bruno Gadelha, Tayana Conte. The best defense is a good defense: adapting negotiation methods for tackling pressure over software project estimates. *ICSE-NIER* , 91-95 (2022).
- Mariam Guizani, Bianca Trinkenreich, Aileen Abril Castro-Guzman, **Igor Steinmacher**, Marco Aurélio Gerosa, Anita Sarma. Perceptions of the State of D&I and D&I Initiative in the ASF. *ICSE-SEIS* , 130-142 (2022).
- Sérgio Ricardo De Oliveira Cavalcante, Bruno Gadelha, Edson Oliveira, **Igor Steinmacher**, Walter Takashi Nakamura, Tayana Conte. Joy to the Work: How Does Software Team Formation Strategy Influence Job Satisfaction? *SBQS* , 15:1-15:10 (2022).
- Mariam Guizani, **Igor Steinmacher**, Jillian Emard, Abrar Fallatah, Margaret Burnett, Anita Sarma. How to Debug Inclusivity Bugs? A Debugging Process with Information Architecture. *ICSE-SEIS* , 90-101 (2022).
- Fabio Santos, Bianca Trinkenreich, João Felipe Pimentel, Igor Wiese, **Igor Steinmacher**, Anita Sarma, Marco Aurélio Gerosa. How to Choose a Task? Mismatches in Perspectives of Newcomers and Existing Contributors. *ESEM* , 114-124 (2022).
- Carlos Danilo Luz, Edson OliveiraJr, Igor Fabio Steinmacher. How Experimentation in Software Engineering has been taught? Survey and Research Agenda. *EASE* , 299-304 (2022).
- Ítalo Santos, Igor Wiese, **Igor Steinmacher**, Anita Sarma, Marco Aurélio Gerosa. Hits and Misses: Newcomers' ability to identify Skills needed for OSS tasks. *SANER* , 174-183 (2022).
- Mairieli Santos Wessel, Ahmad Abdellatif, Igor Wiese, Tayana Conte, Emad Shihab, Marco Aurélio Gerosa, **Igor Steinmacher**. Bots for Pull Requests: The Good, the Bad, and the Promising. *ICSE* , 274-286 (2022).
- Ahmad Abdellatif, Mairieli Santos Wessel, **Igor Steinmacher**, Marco Aurélio Gerosa, Emad Shihab. BotHunter: An Approach to Detect Software Bots in GitHub. *MSR* , 6-17 (2022).
- Bianca Trinkenreich, Ricardo Britto, Marco Aurélio Gerosa, **Igor Steinmacher**. An Empirical Investigation on the Challenges Faced by Women in the Software Industry: A Case Study. *ICSE-SEIS* , 24-35 (2022). ★ Best Paper Award
- Hugo Henrique Fumero de Souza, Igor Wiese, **Igor Steinmacher**, Reginaldo Ré. A characterization study of testing contributors and their contributions in open source projects. *SBES* , 95-105 (2022).
- Felipe Soupinski, Pedro Arantes, **Igor Steinmacher**, Igor Wiese, Hudson Borges, Bruno B. P. Cafeo, Awdren de Lima Fontão. "We are dying!" On Death Signals of Software Ecosystems. *SBES* , 363-369 (2022).
- Edson Dias, Paulo Meirelles, Fernando Castor, **Igor Steinmacher**, Igor Wiese, Gustavo Pinto. What Makes a Great Maintainer of Open Source Projects? *ICSE* , 982-994 (2021). ★ ACM SIGSOFT Distinguished Paper
- Simone de França Tonhão, Thelma Elita Colanzi, **Igor Steinmacher**. Using Real Worked Examples to Aid Software Engineering Teaching. *SBES* , 133-142 (2021).
- Patrícia Matsubara, **Igor Steinmacher**, José Carlos Maldonado, Bruno Gadelha, Tayana Conte. Trust yourself! Or maybe not: factors related to overconfidence and uncertainty assessments of software effort estimates. *SBES* , 452-461 (2021).
- Edson Oliveira Jr., Viviane R. Furtado, Henrique Vignando, Carlos Danilo Luz, André F. R. Cordeiro, **Igor Steinmacher**, Avelino F. Zorzo. Towards Improving Experimentation in Software Engineering. *SBES* , 335-340 (2021).

- Marco Aurélio Gerosa, Igor Wiese, Bianca Trinkenreich, Georg Link, Gregorio Robles, Christoph Treude, **Igor Steinmacher**, Anita Sarma. The Shifting Sands of Motivation: Revisiting What Drives Contributors in Open Source. *ICSE* , 1046-1058 (2021).
- Luiz Philipe Serrano Alves, Igor Scaliante Wiese, Ana Paula Chaves, **Igor Steinmacher**. How to Find My Task? Chatbot to Assist Newcomers in Choosing Tasks in OSS Projects. *CONVERSATIONS* , 90-107 (2021).
- Fabio Santos, Igor Wiese, Bianca Trinkenreich, **Igor Steinmacher**, Anita Sarma, Marco Aurélio Gerosa. Can I Solve It? Identifying APIs Required to Complete OSS Tasks. *MSR* , 346-257 (2021).
- Patrícia G. F. Matsubara, **Igor Steinmacher**, Bruno Gadelha, Tayana Uchôa Conte. Buying time in software development: how estimates become commitments? *CHASE@ICSE* , 61-70 (2021).
- Mairieli Santos Wessel, Alexander Serebrenik, Igor Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. What to Expect from Code Review Bots on GitHub?: A Survey with OSS Maintainers. *SBES* , 457-462 (2020).
- Williamson Silva, Bruno Gadelha, **Igor Steinmacher**, Tayana Conte. Towards an open repository for teaching software modeling applying active learning strategies. *ICSE-SEET* , 162-172 (2020).
- Mairieli Santos Wessel, **Igor Steinmacher**. The Inconvenient Side of Software Bots on Pull Requests. *ICSE (Workshops)* , 51-55 (2020).
- Alexandre Lazaretti Zanatta, Letícia Machado, **Igor Steinmacher**, Rafael Prikladnicki, Cleidson R. B. de Souza. Strategies for Crowdworkers to Overcome Barriers in Competition-based Software Crowdsourcing Development. *ICSE (Workshops)* , 125-128 (2020).
- Luiz Felipe Dias, Caio Barbosa, Gustavo Pinto, **Igor Steinmacher**, Baldoino Fonseca, Márcio Ribeiro, Christoph Treude, Daniel Alencar da Costa. Refactoring from 9 to 5? What and When Employees and Volunteers Contribute to OSS. *VL/HCC* , 1-5 (2020).
- Sogol Balali, Umayal Annamalai, Hema Susmita Padala, Bianca Trinkenreich, Marco Aurélio Gerosa, **Igor Steinmacher**, Anita Sarma. Recommending Tasks to Newcomers in OSS Projects: How Do Mentors Handle It? *OpenSym* , 7:1-7:14 (2020). ★ Best Paper Award
- Marcus Vinicius Bertoncello, Gustavo Pinto, Igor Scaliante Wiese, **Igor Steinmacher**. Pull Requests or Commits? Which Method Should We Use to Study Contributors' Behavior? *SANER* , 592-601 (2020).
- Letícia S. Machado, **Igor Steinmacher**, Sabrina Marczak, Cleidson R. B. de Souza. How Online Forums Complement Task Documentation in Software Crowdsourcing. *ICSE (Workshops)* , 101-108 (2020).
- Mairieli Santos Wessel, Alexander Serebrenik, Igor Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. Effects of Adopting Code Review Bots on Pull Requests to OSS Projects. *ICSME* , 1-11 (2020). ★ IEEE TCSE Distinguished Paper
- James Dominic, Jada Houser, **Igor Steinmacher**, Charles Ritter, Paige Rodeghero. Conversational Bot for Newcomers Onboarding to Open Source Projects. *ICSE (Workshops)* , 46-50 (2020).
- John Anderson, **Igor Steinmacher**, Paige Rodeghero. Assessing the Characteristics of FOSS Contributions in Network Automation Projects. *ICSME* , 324-335 (2020).
- Jefferson De Oliveira Silva, Igor Wiese, Daniel M. Germán, Christoph Treude, Marco Aurélio Gerosa, **Igor Steinmacher**. A theory of the engagement in open source projects via summer of code programs. *ESEC/SIGSOFT FSE* , 421-431 (2020).
- Simone de França Tonhão, Thelma Elita Colanzi, **Igor Steinmacher**. A portal for cataloging worked examples extracted from open source software. *SBES* , 493-498 (2020).
- Giuseppe Iaffaldano, **Igor Steinmacher**, Fabio Calefato, Marco Aurélio Gerosa, Filippo Lanubile. Why do developers take breaks from contributing to OSS projects?: a preliminary analysis. *SoHeal@ICSE* , 9-16 (2019).
- Felipe Fronchetti, Igor Wiese, Gustavo Pinto, **Igor Steinmacher**. What Attracts Newcomers to Onboard on OSS Projects? TL;DR: Popularity. *OSS* , 91-103 (2019).

- Elizamary de Souza Nascimento, Iftekhar Ahmed, Edson Oliveira, Márcio Piedade Palheta, **Igor Steinmacher**, Tayana Conte. Understanding Development Process of Machine Learning Systems: Challenges and Solutions. *ESEM* , 1-6 (2019).
- Adriana Lopes, **Igor Steinmacher**, Tayana Conte. UML Acceptance: Analyzing the Students' Perception of UML Diagrams. *SBES* , 264-272 (2019).
- Gustavo Pinto, Clarice Ferreira, Cleice Souza, **Igor Steinmacher**, Paulo Meirelles. Training software engineers using open-source software: the students' perspective. *ICSE-SEET* , 147-157 (2019). ★ ACM SIGSOFT Distinguished Paper
- Mairieli Santos Wessel, **Igor Steinmacher**, Igor Wiese, Marco Aurélio Gerosa. Should I stale or should I close?: an analysis of a bot that closes abandoned issues and pull requests. *BotSE@ICSE* , 38-42 (2019).
- Fabio Pacheco, Igor Wiese, **Igor Steinmacher**, Bruno Cartaxo, Gustavo Pinto. How Open is the SBES PC Community? *SBES* , 7-11 (2019).
- Gustavo Pinto, Luiz Felipe Dias, **Igor Steinmacher**. Who gets a patch accepted first?: comparing the contributions of employees and volunteers. *CHASE@ICSE* , 110-113 (2018).
- Clarice Ferreira, Cleice Souza, Gustavo Pinto, **Igor Steinmacher**, Paulo Meirelles. When students become contributors: leveraging OSS contributions in software engineering courses. *SBES* , 260-269 (2018).
- Williamson Silva, Bruno Gadelha, **Igor Steinmacher**, Tayana Conte. What are the differences between group and individual modeling when learning UML? *SBES* , 308-317 (2018).
- Gustavo Pinto, **Igor Steinmacher**, Marco Aurélio Gerosa. Leaving Behind the Software History When Transitioning to Open Source: Reasons and Implications. *OSS* , 50-60 (2018).
- Maurício Finavaro Aniche, Christoph Treude, **Igor Steinmacher**, Igor Wiese, Gustavo Pinto, Margaret-Anne D. Storey, Marco Aurélio Gerosa. How modern news aggregators help development communities shape and share knowledge. *ICSE* , 499-510 (2018).
- Alexandre Lazaretti Zanatta, Letícia Machado, **Igor Steinmacher**. Competence, Collaboration, and Time Management: Barriers and Recommendations for Crowdworkers. *CSI-SE@ICSE* , 9-16 (2018).
- Luis Vaz, Sabrina Marczak, **Igor Steinmacher**. An empirical study on task documentation in software crowdsourcing: the case of the topcoder platform. *SBES* , 62-71 (2018). ★ Best Paper Award
- **Igor Steinmacher**, Gustavo Pinto, Igor Scaliante Wiese, Marco Aurélio Gerosa. Almost there: a study on quasi-contributors in open source software projects. *ICSE* , 256-266 (2018).
- Carolina Toscani, Daniel Gery, **Igor Steinmacher**, Sabrina Marczak. A Gamification Proposal to Support the Onboarding of Newcomers in the FLOSScoach Portal. *IHC* , 1:1-1:10 (2018).
- Guilherme C. Diniz, Marco Aurélio Graciotto Silva, Marco Aurélio Gerosa, **Igor Steinmacher**. Using Gamification to Orient and Motivate Students to Contribute to OSS Projects. *CHASE@ICSE* , 36-42 (2017).
- Gustavo Henrique Lima Pinto, Fernando Figueira Filho, **Igor Steinmacher**, Marco Aurélio Gerosa. Training Software Engineers Using Open-Source Software: The Professors' Perspective. *CSEE&T* , 117-121 (2017).
- Jefferson De Oliveira Silva, Igor Scaliante Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa. Students' Engagement in Open Source Projects: An Analysis of Google Summer of Code. *SBES* , 224-233 (2017).
- Williamson Alison Freitas Silva, Igor Fabio Steinmacher, Tayana Uchôa Conte. Is It Better to Learn from Problems or Erroneous Examples? *CSEE&T* , 222-231 (2017).
- Jefferson De Oliveira Silva, Igor Scaliante Wiese, Daniel M. Germán, Igor Fabio Steinmacher, Marco Aurélio Gerosa. How Long and How Much: What to Expect from Summer of Code Participants? *ICSME* , 69-79 (2017).
- Igor Scaliante Wiese, José Teodoro da Silva, **Igor Steinmacher**, Christoph Treude, Marco Aurélio Gerosa. Who is Who in the Mailing List? Comparing Six Disambiguation Heuristics to Identify Multiple Addresses of a Participant. *ICSME* , 345-355 (2016).
- Anita Sarma, Marco Aurélio Gerosa, **Igor Steinmacher**, Rafael Leano. Training the future workforce through task curation in an OSS ecosystem. *SIGSOFT FSE* , 932-935 (2016).

- **Igor Steinmacher**, Tayana Uchôa Conte, Christoph Treude, Marco Aurélio Gerosa. Overcoming open source project entry barriers with a portal for newcomers. *ICSE* , 273-284 (2016).
- Gustavo Pinto, **Igor Steinmacher**, Marco Aurélio Gerosa. More Common Than You Think: An In-depth Study of Casual Contributors. *SANER* , 112-123 (2016).
- Elizamary Nascimento, Williamson Silva, Tayana Conte, **Igor Steinmacher**, Jobson L. Massollar, Guilherme Horta Travassos. Is a Picture worth a Thousand Words?: A Comparative Analysis of Using Textual and Graphical Approaches to Specify Use Cases. *SBES* , 93-102 (2016).
- Luiz Felipe Dias, **Igor Steinmacher**, Gustavo Pinto, Daniel Alencar da Costa, Marco Aurélio Gerosa. How Does the Shift to GitHub Impact Project Collaboration? *ICSME* , 473-477 (2016).
- **Igor Steinmacher**, Tayana Uchôa Conte, Marco Aurélio Gerosa. Understanding and Supporting the Choice of an Appropriate Task to Start with in Open Source Software Communities. *HICSS* , 5299-5308 (2015).
- **Igor Steinmacher**, Tayana Conte, Marco Aurélio Gerosa, David F. Redmiles. Social Barriers Faced by Newcomers Placing Their First Contribution in Open Source Software Projects. *CSCW* , 1379-1392 (2015).
- Igor Scaliante Wiese, Reginaldo Ré, **Igor Steinmacher**, Rodrigo Takashi Kuroda, Gustavo Ansaldi Oliva, Marco Aurélio Gerosa. Predicting Change Propagation from Repository Information. *SBES* , 100-109 (2015).
- Jacilane de Holanda Rabelo, Edson César Cunha de Oliveira, Davi Viana dos Santos, Luis Carlos da Silva Braga, Gleison dos Santos Souza, Igor Fabio Steinmacher, Tayana Uchôa Conte. Knowledge Management and Organizational Culture in a Software Organization - A Case Study. *CHASE@ICSE* , 89-92 (2015).
- **Igor Steinmacher**, Igor Wiese, Tayana Uchôa Conte, Marco Aurélio Gerosa. Increasing the Self-Efficacy of Newcomers to Open Source Software Projects. *SBES* , 160-169 (2015).
- Ricardo T. Geraldi, Edson Oliveira Jr., Tayana Conte, **Igor Steinmacher**. Checklist-based Inspection of SMarty Variability Models - Proposal and Empirical Feasibility Study. *ICEIS* , 268-276 (2015).
- **Igor Steinmacher**, Igor Scaliante Wiese, Tayana Conte, Marco Aurélio Gerosa, David F. Redmiles. The hard life of open source software project newcomers. *CHASE* , 72-78 (2014).
- Igor Scaliante Wiese, Filipe Roseiro Côgo, Reginaldo Ré, **Igor Steinmacher**, Marco Aurélio Gerosa. Social metrics included in prediction models on software engineering: a mapping study. *PROMISE* , 72-81 (2014).
- **Igor Steinmacher**, Ana Paula Chaves, Tayana Uchôa Conte, Marco Aurélio Gerosa. Preliminary Empirical Identification of Barriers Faced by Newcomers to Open Source Software Projects. *SBES* , 51-60 (2014).
- **Igor Steinmacher**, Marco Aurélio Gerosa. How to Support Newcomers Onboarding to Open Source Software Projects. *OSS* , 199-201 (2014).
- **Igor Steinmacher**, Marco Aurélio Gerosa. Choosing an Appropriate Task to Start with in Open Source Software Communities: A Hard Task. *CRIWG* , 349-356 (2014).
- **Igor Steinmacher**, Marco Aurélio Graciotto Silva, Marco Aurélio Gerosa. Barriers Faced by Newcomers to Open Source Projects: A Systematic Review. *OSS* , 153-163 (2014).
- **Igor Steinmacher**, Igor Scaliante Wiese, Ana Paula Chaves, Marco Aurélio Gerosa. Why do newcomers abandon open source software projects? *CHASE@ICSE* , 25-32 (2013).
- Gustavo Ansaldi Oliva, **Igor Steinmacher**, Igor Scaliante Wiese, Marco Aurélio Gerosa. What can commit metadata tell us about design degradation? *IWPSE* , 18-27 (2013).
- Vaninha Vieira, Ana Carolina Salgado, Patrícia C. A. R. Tedesco, Valéria Cesário Times, Carlos Ferraz, Elisa H. M. Huzita, Ana Paula Chaves, **Igor Steinmacher**. The UbiBus Project: Using Context and Ubiquitous Computing to build Advanced Public Transportation Systems to Support Bus Passengers. *SBSI* , 507-512 (2012).
- **Igor Steinmacher**, Igor Scaliante Wiese, Marco Aurélio Gerosa. Recommending mentors to software project newcomers. *RSSE@ICSE* , 63-67 (2012).
- André Luís Schwerz, Rafael Liberato, Igor Scaliante Wiese, **Igor Steinmacher**, Marco Aurélio Gerosa, João Eduardo Ferreira. Prediction of Developer Participation in Issues of Open Source Projects. *SBSC* , 109-114 (2012).

- **Igor Steinmacher**, Igor Scaliante Wiese, Ana Paula Chaves, Marco Aurélio Gerosa. Newcomers Withdrawal in Open Source Software Projects: Analysis of Hadoop Common Project. *SBSC* , 65-74 (2012).
- José Teodoro da Silva, Marco Aurélio Gerosa, Igor Scaliante Wiese, Reginaldo Ré, **Igor Steinmacher**. An Extensible Service for Experts Recommendation on Distributed Software Development Projects. *ICGSE Workshops* , 18-21 (2012).
- Marcelo Koti Nishi, Ana Paula Chaves, **Igor Steinmacher**. A Collective Intelligence Based System for Visualizing Problems in Public Roads. *SBSC* , 94-98 (2012).
- **Igor Steinmacher**, Ana Paula Chaves, Marco Aurélio Gerosa. Awareness Support in Global Software Development: A Systematic Review Based on the 3C Collaboration Model. *CRIWG* , 185-201 (2010).
- Ana Paula Chaves, Elisa Hatsue Moriya Huzita, Vaninha Vieira, **Igor Steinmacher**. A Context Conceptual Model for a Distributed Software Development Environment. *SEKE* , 437-442 (2010).
- **Igor Steinmacher**, José Valdeni de Lima, Elisa Hatsue Moriya Huzita. Task Anticipation: A Quantitative Analysis Using Workflow Process Simulation. *SEKE* , 601-606 (2006).
- **Igor Steinmacher**, Éderson Fernando Amorim, Flávio Luiz Schiavoni, Elisa Hatsue Moriya Huzita. GeCA: Uma Ferramenta de Engenharia Reversa e Geração Automática de Código. *SBSI* , 58-65 (2006).

Supervision

Ongoing Ph.D.

- Pedro A. Oliveira (2024–current), Northern Arizona University — open source governance as a foundation for project sustainability
- Jacob McAuley Penney (2023–current), Northern Arizona University — LLMs to support computational thinking and developer learning
- Morgan Nicholson (2025–current), Northern Arizona University — conversational agents in developer support

Concluded Ph.D.

- Ana Claudia Maciel (2026). Ph.D., Universidade Estadual de Maringá. *Investigating the Impact of GitHub Discussions on Maintainers' Workload and the Dynamics of Open Source Software Communities.* Supervisor.
- Italo Santos (2024). Ph.D., Northern Arizona University. *Supporting Diverse Problem-Solving Styles in Understanding How to Contribute to Open Source.* Co-supervisor (with Marco Gerosa).
- Simone de França Tonhão (2023). Ph.D., Universidade Estadual de Maringá. Supervisor.
- Fabio Santos (2023). Ph.D., Northern Arizona University. Co-supervisor (with Marco Gerosa).
- Marcia Lima (2023). Ph.D., Universidade Federal do Amazonas. Co-supervisor (with Tayana Conte and Bruno Gadelha).
- Bianca Trinkenreich (2022). Ph.D., Northern Arizona University. Supervisor.
- Patricia Matsubara (2022). Ph.D., Universidade Federal do Amazonas. Co-supervisor (with Tayana Conte).
- Mairieli Wessel (2021). Ph.D., Universidade de São Paulo. Co-supervisor (with Marco Gerosa).
- Williamson Silva (2020). Ph.D., Universidade Federal do Amazonas. Co-supervisor (with Tayana Conte).
- Jefferson Silva (2019). Ph.D., Universidade de São Paulo. Co-supervisor (with Marco Gerosa).
- Alexandre Lazaretti Zanatta (2018). Ph.D., Pontifícia Universidade Católica do Rio Grande do Sul. Co-supervisor (with Rafael Prikladnicki).

Concluded M.Sc.

- Misan Etchie (2024–2026). M.Sc., Northern Arizona University. *Improving Learning Outcomes Through a Gamified Software Engineering Learning Environment.* Supervisor.

- Pawan Acharya (2025). M.Sc., Northern Arizona University. *Exploring the Impact of LLM-Supported Collaboration on Team Dynamics and Participant Perceptions in Hackathons.* Supervisor.
- Jacob McAuley Penney (2024). M.Sc., Northern Arizona University. *Using Large Language Models to Support the Computational Thinking Learning Process.* Supervisor.
- Simone de França (2021). M.Sc., Universidade Estadual de Maringá. Supervisor.
- Carlos Danilo Luz (2021). M.Sc., Universidade Estadual de Maringá. Co-supervisor (with Edson Oliveira Jr.).
- Marcus Vinicius Bertoncello (2019). M.Sc., Universidade Estadual de Maringá. Supervisor.
- Anderson Bergamini de Neira (2018). M.Sc., Universidade Estadual de Maringá. Supervisor.
- Carolina Toscani (2018). M.Sc., Pontifícia Universidade Católica do Rio Grande do Sul. Co-supervisor (with Sabrina Marczak).
- João Choma Neto (2017). M.Sc., Universidade Estadual de Maringá. Co-supervisor (with Thelma Colanzi).

Keynotes, Invited Talks and Panels

- **2026** — Keynote, ICSE 2026 Doctoral Symposium (Rio de Janeiro, Brazil)
- **2026** — Panelist, ICSE 2026 Panels and Special Sessions: “SE.next: Raising the Agentic Engineer” (Rio de Janeiro, Brazil)
- **2026** — Invited talks, GiraEuropa 2026, university seminar tour (Italy & Switzerland)
- **2025** — Talk, FOSSY 2025, Science of Community track: “Lessons from a Decade of Open Source Sustainability Research” (Portland, OR, USA)
- **2025** — Panelist, BotSE 2025 (Workshop on Bots in Software Engineering) at ICSE 2025 (Ottawa, Canada)
- **2024** — Keynote, WBots 2024 (Workshop sobre Bots na Engenharia de Software) at CBSoft 2024 (Curitiba, Brazil)
- **2024** — Invited talk, FSE 2024 New Faculty Symposium: “Building Bridges for Sustainable Research” (Porto de Galinhas, Brazil)
- **2024** — Panelist, FSE 2024 Doctoral Symposium: “Life During PhD” (Porto de Galinhas, Brazil)
- **2023** — Keynote, Latin American School of Software Engineering (ELA-ES 2023) at CBSoft 2023, on research methods in empirical software engineering (Campo Grande, Brazil)
- **2019** — Invited talk, BRIDGES 2019 (International Workshop on BRIdging Divides with Globally Engineered Software): “Diversity and Social Aspects of Open Source Software Communities” (Port Moresby, Papua New Guinea)

Professional Service

Journal editorship

- Editorial Board — Empirical Software Engineering (since 2022)
- Editorial Board — Journal of Software: Evolution and Process (since 2021)
- Lead Guest Editor — SpringerOpen JISA, Thematic Series on Free and Open Source Software Development
- Co-Guest Editor — IEEE Software Thematic Series: 20 Years of Open Source

Conference organization

- Student Mentoring Workshop @ ICSE (SMeW 2027)
- General Chair — International Conference on Mining Software Repositories (MSR 2026)
- General Chair — IEEE International Conference on Software Maintenance and Evolution (ICSME 2024)
- General Co-Chair — International Conference on Program Comprehension (ICPC 2024)

- General Chair — International Conference on Cooperative and Human Aspects of Software Engineering (CHASE 2023)
- Virtualization Co-Chair — ICSSP-ICGSE 2021
- Co-organizer — Workshop on Cooperative and Human Aspects of Software Engineering (CHASE 2019)
- Organizer — AZ Computing Postdocs Best Practices (2017)
- General Co-Chair — Brazilian Conference on Software: Theory and Practice (CBSoft 2016)
- Co-organizer — Workshop sobre Aspectos Sociais, Humanos e Econômicos de Software (WASHES 2016)
- General Chair — Free/Libre Open Source Software Workshop (WSL 2014, 2015)
- Organizing Committee — Brazilian Symposium on Collaborative Systems (SBSC 2012)

Program committee chair

- Area Chair, Human Aspects — ICSE 2025
- Program Co-Chair — Brazilian Symposium on Software Engineering, Main Track (SBES 2024)
- Program Co-Chair — Brazilian Symposium on Software Engineering, Insightful Ideas (SBES IIER 2023)
- Program Co-Chair — 18th IFIP Conference on Open Source Systems (OSS 2022)
- Program Co-Chair — International Conference on Software and Systems Process (ICSSP 2022)
- Program Co-Chair — Brazilian Symposium on Software Engineering, Tools Track (SBES Tools 2022)
- Co-Chair — ICSE Student Contest on Software Engineering (SCORE 2021)
- Program Co-Chair — ACM/IEEE International Conference on Global Software Engineering (ICGSE 2020)
- Program Co-Chair — ACS/IEEE International Conference on Computer Systems and Applications (AICCSA 2019)

Program committee membership

- ICSE: International Conference on Software Engineering (2022, 2023, 2024, 2025)
- MSR: International Conference on Mining Software Repositories (2023, 2024)
- EASE: International Conference on Evaluation and Assessment in Software Engineering (2017, 2018, 2024)
- GROUP: ACM International Conference on Supporting Group Work (2020)
- ICSE-SEIS: Software Engineering in Society track (2020)
- SANER Tool Track: International Conference on Software Analysis, Evolution and Reengineering (2019, 2020)
- BotSE: International Workshop on Bots for Software Engineering (2019, 2020, 2022)
- CSCW: ACM Conference on Computer Supported Cooperative Work and Social Computing (2019, 2020)
- ICSE-NIER: New Ideas and Emerging Results track (2019)
- ICGSE: IEEE International Conference on Global Software Engineering (2016, 2017, 2018, 2019)
- SBES: Brazilian Symposium on Software Engineering (2017, 2018, 2019)
- OSS: International Conference on Open Source Systems (2017, 2018)
- ICSME Artifacts Track (2017)
- CHASE: International Workshop on Cooperative and Human Aspects of Software Engineering (2017, 2018)
- CIbSE: Iberoamerican Congress of Software Engineering (2016, 2017)
- SBSC: Brazilian Symposium on Collaborative Systems (2015, 2016, 2017)
- VEM: Workshop on Software Visualization, Evolution, and Maintenance (2015, 2016, 2017)

Journal reviewing

- IEEE Transactions on Software Engineering; ACM TOSEM; ACM Computing Surveys; Communications of the ACM; Empirical Software Engineering; Information and Software Technology; Journal of Systems and Software; Journal of Software: Evolution and Process; Computer Supported Cooperative Work; Computer Standards & Interfaces; Journal of the Brazilian Computer Society; IET Software
- Distinguished Reviewer Award — ICSE 2023

Other service

- Associate Director for Graduate Programs and Academic Affairs — SICCS, NAU
- Chair, proposal committee for NAU's BS in Artificial Intelligence (starts Fall 2026)
- Proceedings Chair — ICSME 2022
- Publicity Chair — ICSME 2018
- Publicity Chair for South America — OSS 2018
- Publicity and Social Media Chair — ICGSE 2017
- Proceedings Chair — ICGSE 2016
- Member of the Education Committee — Brazilian Computer Society (2015–2017)
- Member of the Collaborative Systems Special Interest Group (CESC) — Brazilian Computer Society (2015–2016)

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